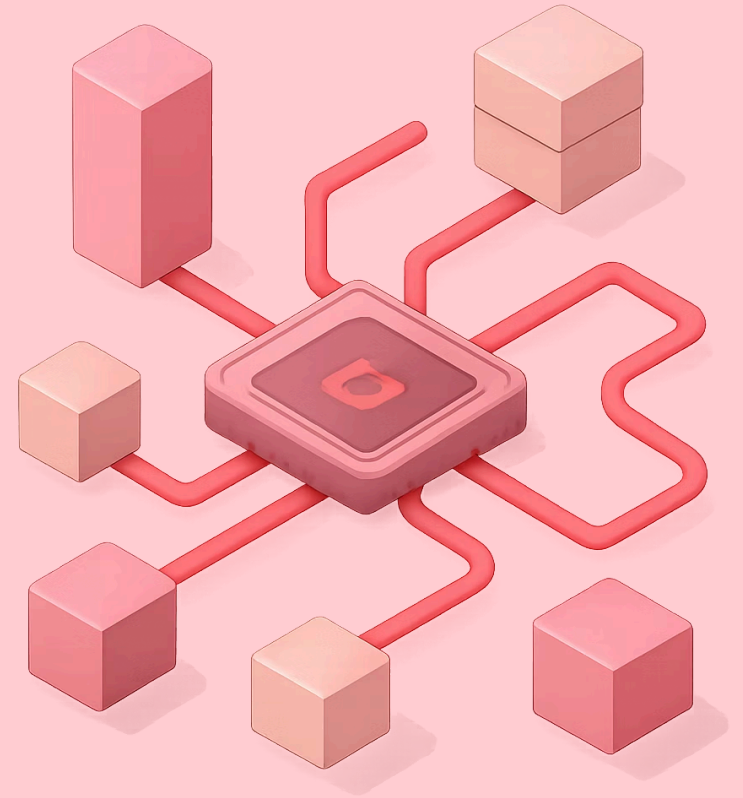


Prompt Engineering



Introduction to Prompt Engineering

Core Concepts

Generative AI

AI systems that generate new content such as text, code, images, audio, or video based on patterns learned from data.

Large Language Model (LLM)

A neural network trained on large amounts of text data to understand and generate language by predicting likely next tokens.

Prompt

A text instruction or query given to an AI model to guide its response.

Prompt Engineering

The practice of crafting effective prompts to elicit desired outputs from AI models.

Why It Matters

Output Quality

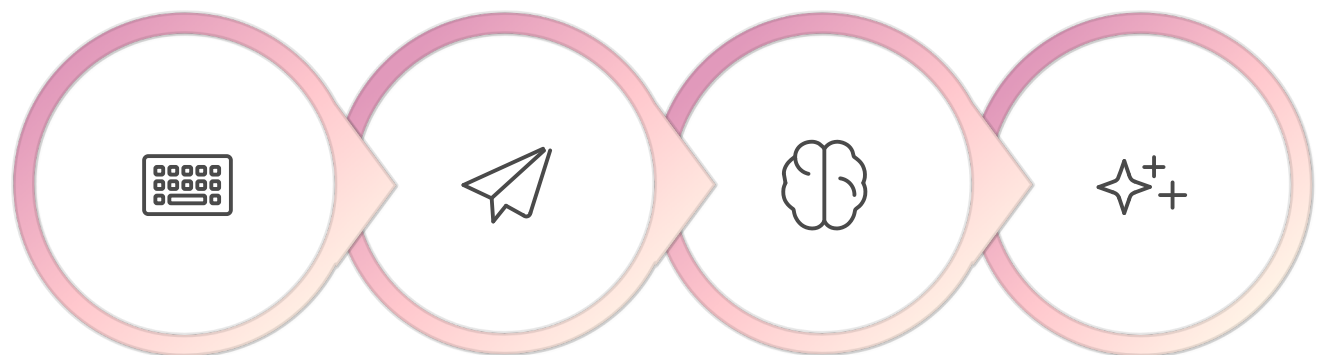
Small changes in wording significantly impact output quality and accuracy.

No Retraining Required

Prompt engineering can improve how an existing model is used without changing or retraining its underlying parameters.

Developer Productivity

Reduces iteration time and is a critical skill in modern software workflows.



User Prompt

Send to LLM

LLM
Processing

AI Output

Anatomy of a Good Prompt

Key Components

Role

"You are a senior code reviewer..."

Task

Clearly state what you want the AI to do.

Context

Provide relevant background and constraints.

Instructions

Specify rules, format, length, or style.

Examples

Show input-output pairs (few-shot prompting).

Output Format

Specify structure: JSON, markdown, bullets, etc.

Poor vs. Engineered Prompt

✗ Poor Prompt

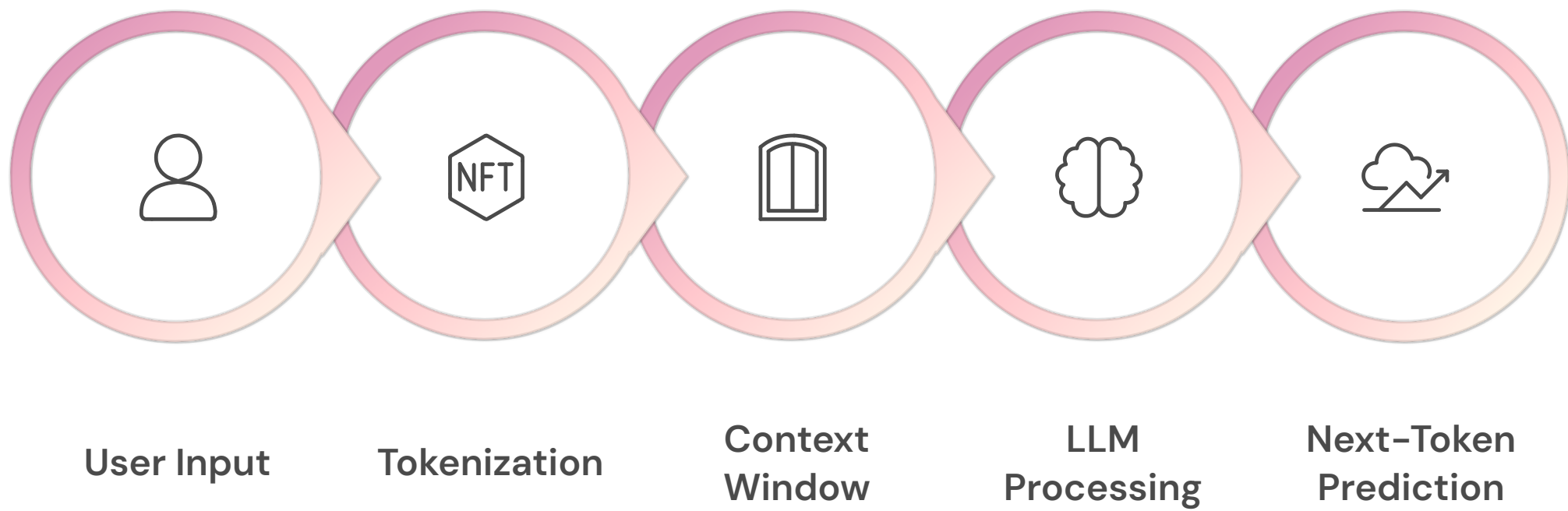
"Fix my code."

✓ Engineered Prompt

"You are a Python expert. Review this function for bugs and performance issues. Provide output as: [Issue], [Severity: High/Medium/Low], [Fix]. Only report actual errors, not style suggestions."

- ❏ Structured prompts produce consistent, actionable, and higher-quality responses.

How Prompting Works & Major Techniques



Zero-shot

Ask the model directly without any examples — relies on pre-trained knowledge.



Few-shot

Provide one or more input-output examples to guide and calibrate model behavior.



Role Prompting

Assign a persona to the model — "You are a DevOps engineer with 10 years experience."



Task Decomposition

Break complex tasks into smaller, sequential steps for better accuracy.



RAG

Retrieval-Augmented Generation — combine external knowledge retrieval with generation.



Tool Calling / Agents

Enable models to call functions or APIs to extend their capabilities dynamically.

Prompt Engineering in Software Development

Practical Applications

→ Requirements & User Stories

Generate, refine, and validate requirements from natural language descriptions.

→ Code Generation & Refactoring

Produce boilerplate, optimize existing code, and suggest architectural improvements.

→ Testing

Auto-generate unit tests, integration tests, and edge-case scenarios.

→ Debugging & Code Review

Root cause analysis, quality assessment, and security vulnerability detection.

→ Documentation & Queries

Write API specs, SQL queries, schema designs, and technical documentation.



Critical Principle: AI assistance does NOT replace testing, code review, or human validation. All AI-generated outputs must be thoroughly reviewed before deployment.

Limitations, Security & Conclusion

Key Limitations & Risks

**Hallucinations**

Models generate plausible-sounding but factually false information.

**Non-Determinism**

Same prompt may produce different results across runs.

**Context Limits**

Models have a maximum context capacity, and inputs that exceed it may be truncated, rejected, or otherwise handled differently depending on the system.

**Prompt Injection**

Malicious or untrusted input attempts to manipulate the model into ignoring or conflicting with its intended instructions.

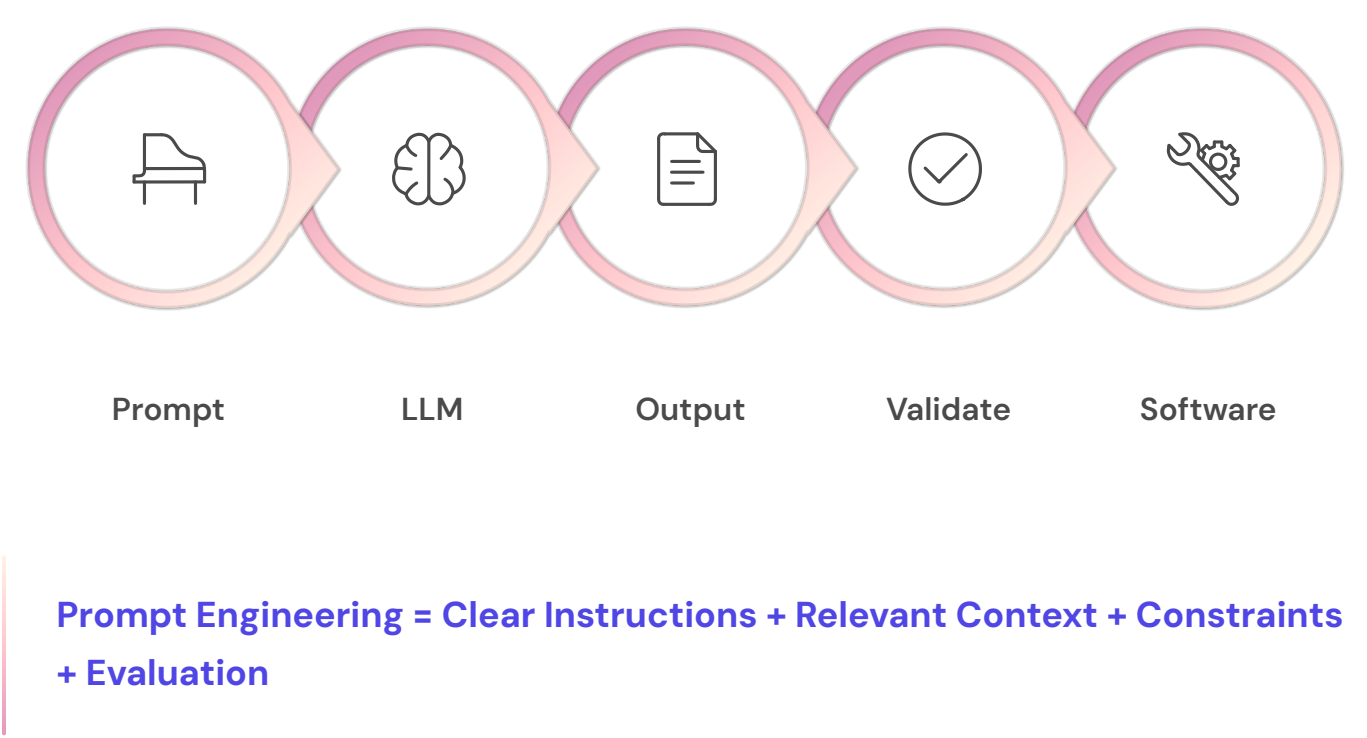
**Data Leakage**

Sensitive data in prompts may be logged or exposed. Never include secrets or PII.

**Jailbreaking**

Adversarial prompts can bypass safety guidelines.

Defensive Practices



01

Validate All Outputs

Always test and review AI-generated code before deployment.

02

Protect Sensitive Data

Never include credentials, API keys, or PII in prompts.

03

Least Privilege

Apply minimal permissions to any AI-connected system or agent.

04

Treat External Input as Untrusted

Content retrieved externally may be adversarially crafted.